

Four-Locus Evolution Feed as Paper 3's Fourth Architectural Claim

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Paper 3 of the CKS theory series commits, as its fourth architectural claim, to a four-locus mechanism that specifies how Full Aspect Integration (FAI) events feed into each participating Self's home evolution machinery. This note formalizes Claim 4 precisely, covers each of its four architectural commitments in detail, states the substrate-ingestion connection that makes the architecture coherent, positions the claim relative to Paper 2's three evolution mechanisms, names four failure modes the claim explicitly defends against, maps five derived sub-commitments (D1.17–D1.21), and provides an operational test. The central framing result is that the four-locus mechanism is not fresh evolutionary architecture: it is what Paper 2's three evolution mechanisms produce when fed by FAI events at the inter-Self perimeter. The fourth locus — the FAI hand-off boundary — is a new input channel to existing mechanisms, not a new mechanism. The instinct-layer non-crossing commitment is treated throughout as an active architectural commitment rather than an absence: Paper 3 explicitly commits that LLM weights and instinct-layer content do not cross the inter-Self perimeter, preserving Paper 2's instinct/reasoning separation at inter-Self scope.

1. The Claim and Its Position in Paper 3

Paper 3 of the CKS theory series establishes six named architectural claims governing inter-Self coordination through a shared substrate. Claim 4 is the fourth of these and is formalized in Paper 3 §7. Its position in the paper's architecture is downstream of three prior claims: Claim 1 (the shared substrate as the architectural object of inter-Self coordination), Claim 2 (Full Aspect Integration as the canonical operation over that substrate), and Claim 3 (the three-tier conflict-handling mechanism at the inter-Self scope). Claim 4 receives those results as inputs and asks what they produce for the evolution of each participating Self after an FAI event closes.

The claim's short form: FAI events feed into each participating Self's home Paper 2 evolution mechanisms through a four-locus mechanism at the FAI hand-off boundary, with substrate ingestion as the connection between the inter-Self event and the home evolution machinery.

The architectural register is *principles-extending*. Two failure modes operate against this calibration. The under-claim would treat Claim 4 as nothing more than a re-citation of Paper 2's three evolution mechanisms with FAI named as an additional input source — dissolving the four loci of fresh content into trilogy-continuity bookkeeping. The over-claim would treat the four loci as fresh evolutionary architecture that competes with Paper 2 §7 rather than extending it — inflating the four loci into a free-standing mechanism decoupled from its inheritance. The precise statement of the claim holds both errors off simultaneously: the four loci contain fresh content that the inheritance does not supply, and everything else is inheritance the claim articulates rather than commitments the claim introduces.

2. The Four Architectural Commitments

2.1 The Hand-Off Mechanism at FAI Dissolution

When the shared substrate dissolves at the close of an FAI event, content propagates to each participating Self's home substrate per governance-configured ingestion at the home perimeter. Three properties characterize this hand-off. The dissolution event triggers ingestion under home governance: the closing of the FAI event is the moment at which the shared substrate transitions from inter-Self scope to home scope, and it is this transition that activates the ingestion machinery. What each home perimeter takes is configurable per home authority: no external party — including the other participating Self or Selves — determines what a given Self's home governance receives from the dissolution. And the absorption itself uses each Self's existing Paper 2 evolution mechanisms operating at home: the hand-off does not install new evolution machinery; it delivers content into machinery that was already operating.

This is the architecturally decisive property of the hand-off locus. The shared substrate dissolves; the home perimeter receives; the home evolution mechanisms process. The hand-off is a moment of substrate transition under governance, not a moment of mechanism transfer or external modification.

2.2 The Per-Mechanism Feed Structure

FAI-derived content reaches Paper 2's three evolution mechanisms at each home perimeter through the existing per-mechanism governance shapes Paper 2 §8 specifies. Those shapes are: verification-substrate machinery for instinct integration, authority architecture for DNA evolution, and proposal-and-acceptance machinery for action-feedback's substrate-change cycles. These shapes do not change when FAI-derived content enters them. The mechanisms operate unchanged; FAI-feed is an additional input source rather than a replacement or a modification of the mechanisms' internal logic.

This is the non-intrusion property of the four-locus mechanism. FAI does not redesign the home evolution machinery. It produces content that home governance routes into that machinery through the same channels any other home substrate content uses. The per-mechanism feed structure is the

architectural statement that no special path exists: FAI outputs are substrate content, and substrate content enters evolution mechanisms through the mechanisms' existing governance shapes.

2.3 The Layer-Routing Rule

The layer of FAI-derived content determines which Paper 2 evolution mechanism receives it. This rule has three explicit cases, and the three-case framing is non-trivial architectural specification: it is not merely the statement that “FAI content reaches mechanisms” but the more precise statement that the layer is what determines routing, with distinct architectural meaning for each case.

DNA-layer content from an FAI event feeds DNA evolution operating on the receiving Self's home DNA. Where home governance authorizes it, DNA-layer content contributed by another Self's aspects through the FAI event can be selectively absorbed into the receiving Self's home DNA layer. This absorption operates through Paper 2's directed selection mechanism — a governance event under home governance authority in which the receiving Self's governance determines what DNA content from the shared substrate is a candidate for absorption, what is approved, and what is declined. The selection is neither automatic nor opaque: it is a human-authorized governance event producing a substrate record.

Action-layer content from an FAI event feeds action-feedback evolution operating within the receiving Self's home action layers. FAI event outcomes become new action-layer content in each participating Self's home substrate. From each Self's home perspective, the FAI event is a class of lived experience that produced action-layer records — records that carry provenance identifying their inter-Self origin. Home action-feedback evolution then operates on this content exactly as Paper 2 specifies for any action-layer content. The FAI event does not require a special action-feedback path; it populates the standard path with content that has a distinctive provenance marker.

Instinct-layer content takes no FAI input by architectural commitment. This is the third case of the layer-routing rule, and it must be stated with the same explicitness as the first two cases — it is not an absence, a gap, or an implicit consequence. Paper 3 commits actively that LLM weights and instinct-layer content do not exchange across the FAI perimeter. Instinct evolution within each Self continues per Paper 2 (through LLM upgrades and substrate-platform infrastructure changes) independently of FAI events. This commitment is what extends Paper 2's instinct/reasoning separation to the inter-Self scope. Paper 2 establishes that instinct-layer and reasoning-layer content evolve through separate mechanisms under distinct governance shapes within a Self; Paper 3 carries this separation across the inter-Self perimeter without modification. Any architecture that permitted LLM weight exchange through FAI events would dissolve this separation at the inter-Self boundary — and Claim 4 closes that path explicitly.

2.4 The Asymmetric Ingestion Property

Different participating Selves may take different things from the same FAI event under their respective home governance authorities. There is no architectural requirement of symmetric absorption. Two Selves that participated in the same FAI event may each absorb different subsets of the event's DNA-layer content into their respective home DNA layers, and may each produce

different action-layer records from the same event, depending on what each Self's home governance configures and authorizes.

This is the governance-sovereignty property of the hand-off boundary. Each Self's home governance has sole authority over what its home substrate receives at dissolution. The shared substrate produces a jointly-produced event record; the dissolution produces per-home ingestion under per-home authority. The asymmetric ingestion property is the architectural statement that FAI does not require or produce convergence of the participating Selves' home substrates. They coordinate; they do not merge.

3. The Substrate-Ingestion Connection

The four architectural commitments above share a common mechanism: substrate ingestion. FAI outputs enter each participating Self's home evolution machinery as inputs — as substrate content flowing into mechanisms that were already operating under home governance. This is the architectural fact that makes Claim 4 coherent without being redundant.

The negative form of this commitment is important: there is no direct manipulation of either Self's home-substrate internals through FAI. Evolution from FAI events does not occur by having the FAI event modify a Self's DNA layer or action layer from the outside. It occurs by having the FAI event produce content that enters the home substrate at dissolution, after which the home evolution mechanisms process it as they would any other home substrate content.

This framing — substrate ingestion, not direct modification — is what preserves Paper 2's governance properties for within-home evolution while making FAI productive for that evolution. If FAI events could directly modify a Self's home substrate internals, they would bypass the per-mechanism governance shapes through which Paper 2 §8 organizes evolution authority. They would short-circuit the verification-substrate machinery for instinct integration, the authority architecture for DNA evolution, and the proposal-and-acceptance machinery for action-feedback. Substrate ingestion is the architectural commitment that prevents this bypass. FAI outputs are received at the home perimeter; they enter existing governance channels; home governance retains full authority over what happens to them from that point forward.

There is no "FAI integration API." The architecture reuses existing Paper 2 mechanisms by treating FAI outputs as a new class of home substrate inputs. The novelty lives in the input channel — the hand-off boundary — not in the processing machinery.

4. Relationship to Paper 2's Three Evolution Mechanisms

Paper 2 established three evolution mechanisms in productive tension within each Self at home perimeter: instinct evolution (through LLM version changes and infrastructure updates), DNA evolution (through governance-authored directed selection), and action-feedback evolution (through the proposal-and-acceptance cycle over operational evidence). These three mechanisms apply at all three compositional levels Paper 2 specifies — cell, aspect, and Self — with horizontal and vertical

evolution axes, all under unified human governance.

Claim 4 does not introduce a fourth mechanism to this list. It introduces a fourth *locus* — the FAI hand-off boundary — that serves as a new input channel to the existing mechanisms. The mechanisms themselves are unchanged. The distinction matters for how the claim is read defensively: an architecture that adds a fourth mechanism would need to specify that mechanism's governance shape, its relationship to the other three, its operational preconditions, and its failure modes. Claim 4 does none of this, because nothing of this kind is needed. The input channel has a governance shape (governance-configured ingestion under home authority); the mechanisms it feeds already have their governance shapes (from Paper 2 §8). The claim's fresh content is at the interface between the channel and the mechanisms — the layer-routing rule, the asymmetric ingestion property, the instinct non-crossing commitment — not at the mechanisms themselves.

The most compact statement of the relationship: Paper 2 established three evolution mechanisms within a Self; Paper 3 extends the evolution architecture with a fourth locus at the inter-Self boundary, feeding into the existing Paper 2 mechanisms with the same underlying mechanisms (action-feedback, directed selection) receiving new inputs from FAI events.

5. Four Failure Modes the Claim Defends Against

Claim 4 is partly constituted by what it excludes. Four failure modes identify architectures that differ from the four-locus mechanism in ways that reduce governance, traceability, or evolutionary integrity.

LLM weight exchange as an evolution mechanism. An architecture in which FAI events cause participating Selves to exchange or merge LLM weights would dissolve the instinct/reasoning separation at the inter-Self boundary. The instinct-layer non-crossing commitment closes this path. LLM weight exchange is not evolution-feed at inter-Self scope; it is a different architectural move that bypasses the layer-routing rule entirely and operates outside the governance channels through which Paper 2 organizes evolution authority. Claim 4 does not merely decline to include weight exchange — it explicitly commits to its absence as an architectural property.

Direct manipulation of home-substrate internals. An architecture in which FAI events directly modify a participating Self's DNA layer or action layer from outside the home perimeter would bypass the per-mechanism governance shapes. The substrate-ingestion connection closes this path. Evolution from FAI events must flow through home substrate ingestion at the hand-off boundary, entering the home evolution mechanisms through their existing governance channels. Direct external modification of home-substrate internals is not substrate ingestion; it is a governance violation.

Ungoverned absorption. An architecture in which DNA-layer content from an FAI event automatically enters a receiving Self's home DNA without governance authorization would strip the directed-selection governance shape from the DNA evolution locus. The layer-routing rule, applied to DNA-layer content, routes it to DNA evolution operating under Paper 2's authority architecture for DNA evolution — a governance event under home governance authority, not an automatic merge.

Ungoverned absorption is not a valid instance of the four-locus mechanism.

Unattributed evolution from FAI. An architecture in which action-layer records produced from FAI events carry no provenance identifying their inter-Self origin would make evolution from FAI events indistinguishable from evolution from home events. The action-feedback feed locus carries the architectural commitment that action-layer content from FAI events is attributed to its inter-Self origin. This provenance is what makes inspection possible and what makes the history of each Self's evolution traceable across the FAI perimeter.

6. Derived Sub-Commitments: D1.17–D1.21

Claim 4 decomposes into five foundational sub-commitments at the D1 level. Each sub-commitment is independently formalizable and independently subject to challenge; the five together constitute the full architectural content of the claim.

D1.17 — FAI dissolution hand-off. The dissolution of the shared substrate at the close of an FAI event is the triggering event for governance-configured ingestion into each participating Self's home substrate. The hand-off mechanism specifies that dissolution triggers ingestion, that ingestion is configurable per home authority, and that the ingestion uses existing Paper 2 evolution mechanisms.

D1.18 — Per-mechanism feed structure. FAI-derived content reaches Paper 2's three evolution mechanisms at each home perimeter through the existing per-mechanism governance shapes from Paper 2 §8. The mechanisms operate unchanged. FAI-feed is an additional input source, not a modification of the mechanisms or their governance shapes.

D1.19 — Three-case layer-routing rule. The layer of FAI-derived content determines which Paper 2 evolution mechanism receives it. DNA-layer content routes to DNA evolution. Action-layer content routes to action-feedback evolution. Instinct-layer content routes to no mechanism — it does not cross the inter-Self perimeter. The three-case framing is explicit and exhaustive.

D1.20 — Instinct-takes-no-FAI-input commitment. LLM weights and instinct-layer content do not exchange across the FAI perimeter. This commitment is active rather than silent: Paper 3 explicitly closes the path, extending Paper 2's instinct/reasoning separation to the inter-Self scope. Instinct evolution within each Self continues independently of FAI events through the mechanisms Paper 2 specifies.

D1.21 — Asymmetric ingestion. Different participating Selves may take different things from the same FAI event under their respective home governance authorities. There is no architectural requirement of symmetric absorption. Each Self's home governance has sole authority over what its home substrate receives at FAI dissolution.

7. Operational Test

For a system asserting that FAI event outputs are entering a Self's home substrate through the four-locus mechanism, the following operational test applies.

An observer with access to the home substrate and its governance records should be able to:

1. Find action-layer records attributed to the FAI event's inter-Self origin. Action-layer content produced from FAI event outcomes should carry provenance identifying that it originated from an inter-Self coordination event rather than from purely home operations. The absence of provenance attribution would indicate that action-layer records from FAI events are being ingested without the attribution the four-locus mechanism requires.

2. Confirm that any DNA absorption was authorized through directed selection under home governance. Where the receiving Self's home substrate contains DNA-layer content that originated from another Self's FAI contribution, a governance record should exist documenting that the absorption was authorized through directed selection under home governance authority — that a governance event occurred, that it was human-authorized, and that it operated through the authority architecture Paper 2 specifies for DNA evolution. The absence of such a record would indicate ungoverned absorption, which is not a valid instance of the four-locus mechanism.

3. Verify that no LLM weight exchange occurred. The instinct-layer non-crossing commitment is architecturally verifiable in the negative: the Self's LLM weights after the FAI event should be explicable by instinct evolution mechanisms the Self's home governance authorized — LLM upgrades, infrastructure changes — and not by any exchange with another Self's LLM. A weight change that cannot be traced to a home-governed instinct evolution event would constitute a violation of the non-crossing commitment.

Together, these three checks constitute the operational test for the four-locus mechanism. A system that passes all three is operating in accordance with Claim 4. A system that fails any one of them is operating outside it, in ways the four failure modes in §5 specifically name.

8. Conclusion

Paper 3's fourth architectural claim establishes the four-locus evolution-feed mechanism as what Paper 2's three evolution mechanisms produce when fed by FAI events through the shared substrate at the inter-Self perimeter. The claim's four architectural commitments — the hand-off mechanism at FAI dissolution, the per-mechanism feed structure, the layer-routing rule with its three explicit cases, and the asymmetric ingestion property — together specify how inter-Self coordination becomes productive for each participating Self's home evolution without dissolving the governance properties that make home evolution trustworthy.

The substrate-ingestion connection is the architectural fact that holds this together. FAI outputs enter existing Paper 2 mechanisms as home substrate inputs; no direct manipulation of home-substrate internals occurs; no special integration path is required. The instinct-layer non-crossing commitment is the architectural fact that holds the instinct/reasoning separation across the inter-Self boundary: LLM weights and instinct-layer content do not exchange, and Paper 3 commits to this actively. The asymmetric ingestion property is the architectural fact that holds governance sovereignty at the home perimeter: each Self takes what its own governance authorizes, independently of what other

participating Selves take.

Five derived sub-commitments (D1.17–D1.21) formalize the claim’s decomposition into independently challengeable units. The operational test in §7 provides an observer-accessible verification procedure for each of the three loci that do cross the inter-Self perimeter — action-feedback feed, DNA evolution feed, and the instinct non-crossing commitment.

References

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